

```
mysql> INSERT INTO Student VALUES
-> (101, 'Ravi', 'CSE', 8.50, 20);
Query OK, 1 row affected (0.05 sec)

mysql>
mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+
| RegNo | Name  | Dept  | CGPA  | Age  |
+-----+-----+-----+-----+-----+
| 101   | Ravi  | CSE   | 8.50  | 20   |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

1.

2.

```
mysql> INSERT INTO Student (RegNo, Name, Dept, CGPA, Age) VALUES
-> (102, 'Anu', 'CSE', 8.70, 20),
-> (103, 'Priya', 'ECE', 9.10, 21),
-> (104, 'Kiran', 'CSE', 7.80, 20),
-> (105, 'Rahul', 'ECE', 8.40, 22),
-> (106, 'Sneha', 'IT', 9.20, 20),
-> (107, 'Arjun', 'CSE', 8.90, 21);
Query OK, 6 rows affected (0.05 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM student;
+-----+-----+-----+-----+-----+
| RegNo | Name  | Dept  | CGPA  | Age  |
+-----+-----+-----+-----+-----+
| 101   | Ravi  | CSE   | 8.50  | 20   |
| 102   | Anu   | CSE   | 8.70  | 20   |
| 103   | Priya | ECE   | 9.10  | 21   |
| 104   | Kiran | CSE   | 7.80  | 20   |
| 105   | Rahul | ECE   | 8.40  | 22   |
| 106   | Sneha | IT    | 9.20  | 20   |
| 107   | Arjun | CSE   | 8.90  | 21   |
+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> SELECT Dept, COUNT(*) AS Number_of_Students
-> FROM Student
-> GROUP BY Dept;
+-----+-----+
| Dept  | Number_of_Students |
+-----+-----+
| CSE   | 4                  |
| ECE   | 2                  |
| IT    | 1                  |
+-----+-----+
3 rows in set (0.00 sec)
```

3.

```
ERROR 1054 (42S22): Unknown column 'Student.DeptID' in 'on clause'
mysql>
mysql> SELECT Student.Name, Department.DeptName
-> FROM Student
-> INNER JOIN Department
-> ON Student.Dept = Department.DeptID;
Empty set, 6 warnings (0.03 sec)

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+
| RegNo | Name  | Dept | CGPA | Age |
+-----+-----+-----+-----+-----+
| 101   | Ravi  | CSE  | 8.50 | 20  |
| 102   | Anu   | CSE  | 8.70 | 20  |
| 103   | Priya | ECE  | 9.10 | 21  |
| 104   | Kiran | CSE  | 7.80 | 20  |
| 105   | Rahul | ECE  | 8.40 | 22  |
| 106   | Sneha | IT   | 9.20 | 20  |
| 107   | Arjun | CSE  | 8.90 | 21  |
+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM Department;
+-----+-----+-----+
| DeptID | DeptName | HOD   |
+-----+-----+-----+
| 1      | CSE      | Kiran |
| 2      | ECE      | Arjun |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

4.

```
mysql>
mysql> SELECT Student.RegNo, Student.Name, Student.CGPA,
-> Department.DeptName, Department.HOD
-> FROM Student
-> INNER JOIN Department
-> ON Student.DeptID = Department.DeptID;
+-----+-----+-----+-----+-----+
| RegNo | Name  | CGPA | DeptName | HOD   |
+-----+-----+-----+-----+-----+
| 101   | Asha  | 8.75 | CSE      | Kiran |
| 102   | Ravi  | 9.20 | ECE      | Arjun |
| 103   | Priya | 8.50 | CSE      | Kiran |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

5.

```
mysql> SELECT Student.RegNo, Student.Name, Student.CGPA,
-> Department.DeptName, Department.HOD
-> FROM Student
-> LEFT JOIN Department
-> ON Student.DeptID = Department.DeptID;
+-----+-----+-----+-----+-----+
| RegNo | Name  | CGPA | DeptName | HOD   |
+-----+-----+-----+-----+-----+
| 101   | Asha  | 8.75 | CSE      | Kiran |
| 102   | Ravi  | 9.20 | ECE      | Arjun |
| 103   | Priya | 8.50 | CSE      | Kiran |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```